

Guest Editorial of the Special Section on Advances in Neural Computing-Enabled Device Health Management for Consumer Technology

CONSUMER technology has undergone an evolution from simple manually operated devices to complex intelligent systems that integrate seamlessly into daily life. The rapid proliferation and integration of consumer technology in our daily lives has brought about the need for more elaborate development of health management systems to ensure device reliability, resilience, and longevity. Unexpected failures of consumer technology, including both hardware and software, can lead to significant inconvenience in daily lives and work, financial costs, health and safety risks for users, increased warranty and repair costs, brand reputation damage, and legal issues for manufacturers.

Challenges remain to realize highly adaptive, weak-supervision, efficient, and secure intelligence for consumer technology health management. Neural computing-enabled methods are increasingly being widely applied throughout different aspects of consumer technology. The learning paradigm of neural computing-enabled methods has evolved rapidly in recent years, from supervised learning, and semi-supervised learning to unsupervised and self-supervised learning. Therefore, breakthroughs have been witnessed in various regression, classification, model updating and optimization tasks, supported by different kinds of data such as numeric sensor values, images, speech signals, texts, etc. Advanced neural computing-enabled techniques, incorporated with optimization, control theory, reliability and resilience modeling, etc. are expected to form advanced health management methods for consumer electronics to avoid unexpected failure, extend the life span, and reduce costs for consumers and manufacturers.

Based on the above premises, this Special Section on Advances in Neural Computing-enabled Device Health Management for Consumer Technology focusing on consumer electronics health management, such as early anomaly detection, fault diagnosis, remaining useful life (RUL) prediction, predictive maintenance, usage recommendation, reliability modeling, resilience analysis, personalized usage pattern recommendation, etc. enabled by advanced neural computing-enabled techniques. The Special Section was organized by several representative of the Technical Committees of IEEE Consumer Technology Society in these areas (<https://ctsoc.ieee.org/technical.html>). Overall, 14 submissions were received spanning different consumer electronic (CE)

domains, which were prescreened and reviewed by experts in the fields. At the end of the review process, 2 papers were ultimately accepted.

The work in [A1] focuses on safety to protect patients from potential risks in clinical trials using healthcare consumer electronics. Serious adverse events (SAEs) in clinical trials may pose significant safety threats to patients and incur substantial economic losses. To predict and prevent SAEs, a universal dataset for Serious Adverse Events prediction (SerAE) is built by aggregating trial protocol, drug, and disease information from multiple data sources. The dataset encompasses 10,643 clinical trials, 4,512 diseases, and 2,563 drugs. Furthermore, an Event-aware Dual Representation model with mixture-of-Expert (EDRE) for serious adverse event prediction is proposed to achieve universal prediction of SAEs. The experiments on SerAE demonstrate that EDRE significantly enhances the performance in predicting SAEs compared to state-of-the-art baseline models.

In [A2], a novel backbone network, referred to as multiscale convolutional reservoir computing (MCRC), is proposed to learn features in multiple time-scales for Li-ion battery RUL prediction based on only previous capacity measurements. Li-ion batteries are widely employed as power sources for consumer electronics, whose degradation is complex and contains temporal dependencies in different time-scales. The proposed MCRC employs convolutional filters to extract multiscale temporal features, whose multiscale sustained dependence are further captured by multiple reservoir layers. Benchmark and in-house experiments show that the proposed MCRC can accurately predict RUL of Li-ion batteries, outperforming other state-of-the-art backbone networks.

In general, the above research topics include neural networks, data modeling and prediction for health management. All of them focus on neural computing-enabled technologies on consumer electronics.

APPENDIX: RELATED ARTICLES

- [A1] B. Kan et al., "An event-aware dual representation model with mixture-of-experts for serious adverse events prediction in clinical trials," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 3337–3346, May 2025, doi: [10.1109/TCE.2025.3572452](https://doi.org/10.1109/TCE.2025.3572452).
- [A2] C. Li, G. Guo, and Z. Pu, "Multiscale convolutional reservoir computing for remaining useful life prediction of li-ion batteries," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 3347–3356, May 2025, doi: [10.1109/TCE.2025.3570594](https://doi.org/10.1109/TCE.2025.3570594).

YU WANG, *Guest Editor*

Department of Mechanical Engineering
Xi'an Jiaotong University
Xi'an 710049, China
E-mail: ywang95@xjtu.edu.cn

ZHE YANG, *Guest Editor*

Department of Mechanical Engineering
Dongguan University of Technology
Dongguan 523808, China
E-mail: yangz@dgut.edu.cn

BIN ZHANG, *Guest Editor*

Department of Electrical Engineering
University of South Carolina
Columbia, SC 29208 USA
E-mail: zhangbin@cec.sc.edu

YANXIA SUN, *Guest Editor*

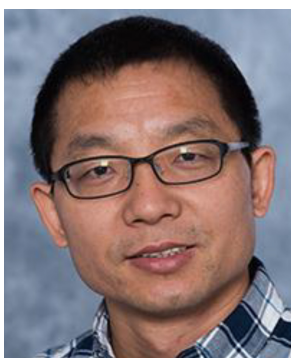
Department of Electrical and Electronic Engineering Science
University of Johannesburg
Johannesburg 2092, South Africa
E-mail: ysun@uj.ac.za



Yu Wang (Senior Member, IEEE) received the B.Eng. degree in mechanical design and manufacturing automation in 2005, the M.Eng. degree in manufacturing engineering and automation in 2008, and the Ph.D. degree in systems engineering and engineering management in 2014. From 2016 to 2018, he was a Postdoctoral Research Fellow with the Department of Mechanical Engineering, University of California at Berkeley, Berkeley, CA, USA. His current research interests include data mining, machine learning, reliability modeling and assessment, fault prognostics, and health management. He has published over 100 technical papers in related fields. He was the Program Co-Chair of several top conferences, including the International Conference on Neural Computing for Advance Applications. He is currently an Associate Editor of the IEEE ACCESS.



Zhe Yang (Member, IEEE) received the B.E. degree in measurement and control and instrument and the M.Sc. degree in mechanical engineering from Xi'an Jiaotong University, Xi'an, China, in 2012 and 2015, respectively, and the Ph.D. degree in energy and nuclear science and technology from the Politecnico di Milano, Milan, Italy, in 2020. From 2020 to 2022, he was a Postdoctoral Research Fellow with the Department of Mechanical Engineering, Xi'an Jiaotong University, Xi'an, China. Since 2022, he has been with the Dongguan University of Technology, Dongguan, China, where he is currently an Associate Professor of Industrial Engineering with the School of Mechanical Engineering. His research interests include prognostics and health management of industrial equipment, machine learning, combinatorial optimization, and intelligent monitoring systems.



Bin Zhang (Senior Member, IEEE) received the B.E. and M.E. degrees in mechanical engineering from the Nanjing University of Science and Technology, Nanjing, China, in 1993 and 1999, respectively, and the Ph.D. degree in electrical engineering from Nanyang Technological University, Singapore, in 2007. He is currently an Associate Professor with the Department of Electrical Engineering, University of South Carolina, Columbia, SC, USA. He was with Research and Development, General Motors, Detroit, MI, USA; Impact Technologies, Rochester, NY, USA; and the Georgia Institute of Technology, Atlanta, GA, USA. His current research interests include prognostics and health management, intelligent systems and controls, and their applications to various engineering systems. He is currently an Associate Editor of IEEE TRANSACTIONS ON AUTOMATION SCIENCE AND ENGINEERING and *Neurocomputing*.



Yanxia Sun (Senior Member, IEEE) received the D.Tech. degree in electrical engineering from the Tshwane University of Technology, South Africa, and the Ph.D. degree in computer science from University Paris-EST, France, in 2012. She is currently a Professor with the Department of Electrical and Electronic Engineering Science, University of Johannesburg, South Africa. Her research interests are deeply rooted in the development of nature-inspired optimization algorithms, multiobjective evolutionary optimization, and the application of deep learning in health and engineering. Her contributions to her field are recognized globally, marked by her inclusion in the World's Top 2% Scientists by citations in 2023. Her leadership extends beyond teaching and research; she actively participates in academic and industry collaborations, contributing to advancements in engineering education and practice internationally.